Cloud Computing & Big Data Infrastructure

Outline

- Clouds can be classified in terms of who owns and manages the cloud;
- Based on a deployment model, we can classify cloud as
  - Public
  - Private
  - Hybrid
  - Community cloud

Cloud Deployment Model

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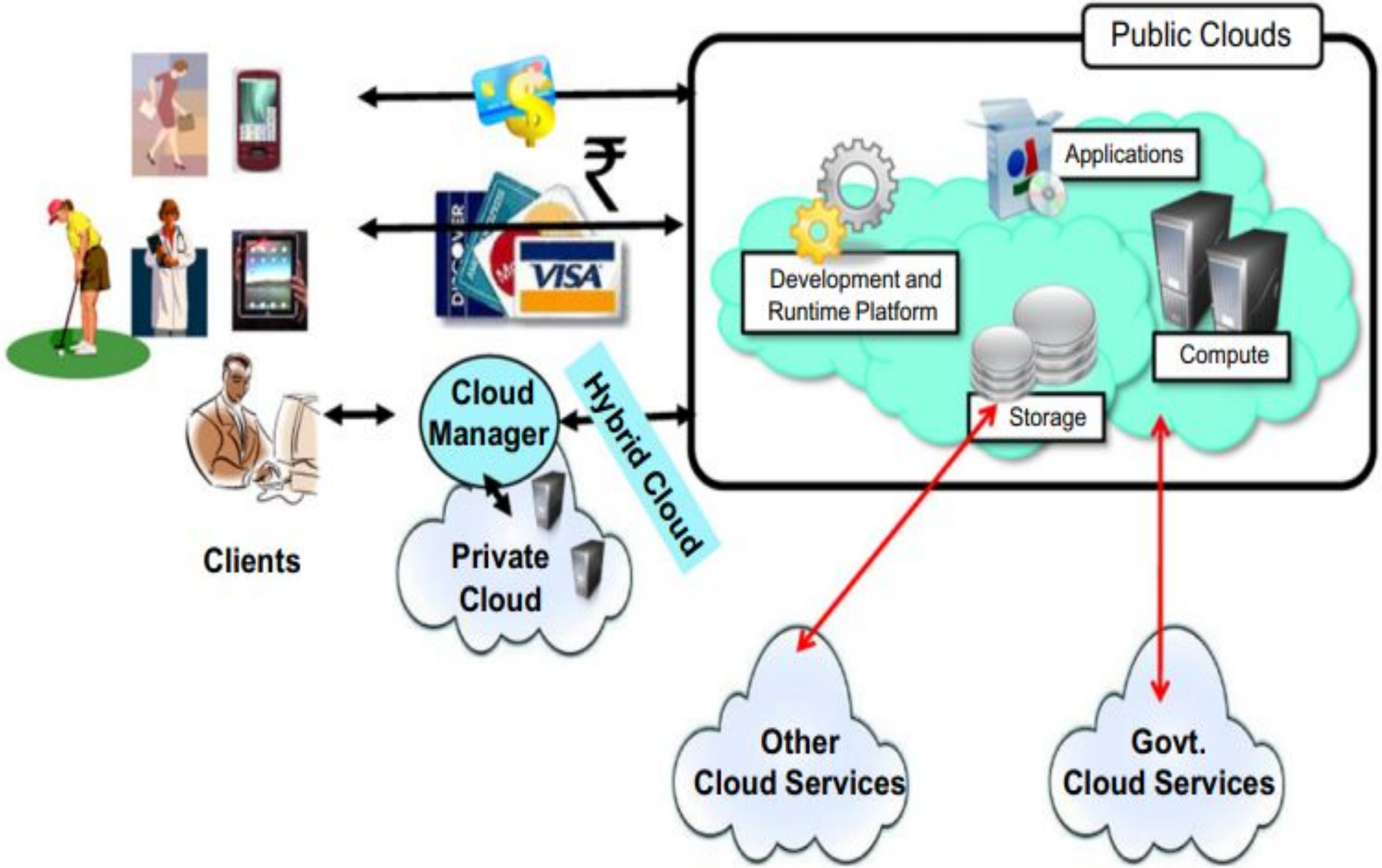
- Based on a deployment model, we can classify cloud as
  - Public
  - Private
  - Hybrid/Community cloud

Cloud Deployment Model



Public Clouds

- First cloud model to be implemented and offered to the public.
- Infrastructure is hosted by third-party cloud providers.
- Services are accessible globally via the Internet, anytime.
- Offers high performance and scalability, but shares resources—raising security concerns.

Public Clouds

- **Ideal for Startups & SMEs:** Offers **cost-effective access** to cloud resources with **no upfront infrastructure investment**.

- **Global Reach with Expert Support:** Providers like **GlobalDots deliver worldwide services** and offer expert guidance to help users select the **most suitable solutions**.

- **Multitenancy by Design:** Enables **multiple customers to share a single software instance**, maximizing efficiency and scalability.

- **Quality of Service (QoS) Management:** Ensures **reliable performance**, resource allocation, and **service-level compliance** in multi-user environments.

Public Clouds

- **Resource Monitoring & Billing:** A significant portion of the cloud software stack is dedicated to **monitoring resource usage**, **generating usage-based bills**, and maintaining a **detailed usage history** for each customer. *Examples:* Amazon EC2 (IaaS), Google App Engine (PaaS)

- **Geo-Distributed Datacenters:** Public cloud providers operate **geographically dispersed datacenters** to balance user load and **optimize latency** based on user location. *Example:* Amazon Web Services (AWS) maintains datacenters in the US, Europe, Singapore, and Australia.

- **Regional Service Options:** Customers can select from multiple regions (e.g., `us-west-1`, `us-east-1`, `eu-west-1`), offering **location-based flexibility** and **differentiated pricing** across regions.

Important Concerns – Public Clouds

- **Sensitive Data Risk:** Not ideal for storing highly confidential data.
- **Loss of Control:** Users lack control over infrastructure and data location.
- **Security Issues:** More prone to breaches due to shared environment.
- **Not for Critical Use:** Unsuitable for government or military applications.
- **Service Abuse:** Can be misused by malicious users or hackers.
- **DoS Attacks:** Susceptible to denial-of-service disruptions.

Private Clouds

- Exclusively operated for a single organization.
- Offers high control, customization, and security.
- Preferred by large enterprises and government bodies.
- Avoids public cloud issues like bandwidth limits and compliance risks.

Benefits: Private Clouds

- Custom hardware/software deployment and configuration.
- Deep visibility into security and access control.
- Enforces internal compliance—no dependency on vendor standards.

Disadvantages: Private Clouds

- High upfront cost.
- Limited scalability.
- Restricted external access.
- Complex setup requiring skilled staff.
- Higher management overhead.

Hybrid Clouds

- Used when private cloud **cannot meet QoS demands.**
- Combines **private infrastructure** with **public cloud resources.**
- Offers flexibility to **scale workloads** as needed.

Benefits: Hybrid Clouds

- Leverages existing IT infrastructure.
- Keeps sensitive data on-premises.
- Scales flexibly using public cloud when needed.
- Enhances reliability through multi-environment setup.
- Concerns: Public-side security & system compatibility.

Community Cloud

- "**Community cloud** is created by integrating the services of different clouds to address the specific needs of an industry, a community, or a business sector".
- NIST Definition "The infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g., mission, security requirements, policy, and compliance considerations)".
- It may be managed by the organizations or a third party and may exist on premise or off premise

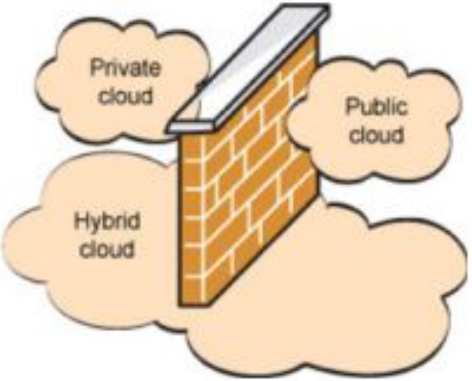
NIST "National Institute of Standards and Technologies"

Community Cloud

- Different organizations such as government bodies, private enterprises, research organizations, and even public virtual infrastructure providers contribute with their resources to build the cloud infrastructure.
- Example Scenarios
  - UIDAI Aadhaar Ecosystem Cloud: A secure community cloud by UIDAI for Aadhaar registrars, authentication agencies, and KYC providers to handle enrolment, authentication, and verification.
  - National Academic Depository (NAD) Cloud: A Ministry of Education–managed cloud integrated with DigiLocker for secure sharing of academic records among institutions.

Cloud Deployment Models

Public/Internet Clouds	Private/Enterprise Clouds	Hybrid/Inter Clouds
*Third-party, multitenant cloud infrastructure and services *Available on a subscription basis to all	*A public cloud model within a company's own datacenter/infrastructure for internal and/or partners' use	* Mixed use of private and public clouds; leasing public cloud services when private cloud capacity is insufficient



Introduction to Cloud Service Models

Cloud service models represent different layers of abstraction in cloud computing. They define what level of control and responsibility the user has and how the cloud resources are delivered. The three foundational models are:

- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)

Each model serves different needs, from providing raw compute power to complete, ready-to-use applications.

Infrastructure as a Service (IAAS)

- IaaS offers fundamental computing resources like virtual machines, storage, and networks on a pay-as-you-go basis.
- **What You Get:**
  - Virtual machines (VMs)
  - Block and object storage
  - Firewalls, load balancers, and IP addresses
- **User Responsibility:**
  - Operating System (OS)
  - Middleware
  - Runtime
  - Applications
  - Data

Infrastructure as a Service (IAAS) (Cont'd)

- Advantages:
  - Full control over infrastructure
  - Scalable and elastic
  - Cost-effective for temporary workloads
- Examples:
  - Amazon EC2: Launch virtual servers with full control over configuration.
  - Google Compute Engine: Run VMs on Google's infrastructure.
  - Microsoft Azure VMs: Deploy Windows or Linux VMs for app hosting.
- Use Cases:
  - Running virtual desktops
  - Hosting websites
  - Backup and recovery systems

Platform as a Service (PaaS)

- PaaS provides a platform allowing customers to develop, run, and manage applications without dealing with infrastructure complexity.
- What You Get:
  - Application hosting environment
  - Development frameworks
  - Databases and caching
  - Monitoring and scaling tools
- User Responsibility:
  - Application code
  - Business logic
  - Data

Platform as a Service (PaaS)

Advantages:

Faster time-to-market
No server management
Built-in scalability and monitoring

Examples:

Google App Engine: Host and scale applications automatically.
Heroku: Easy deployment of web apps using Git.
Microsoft Azure App Services: Web apps, RESTful APIs, and mobile backends.

Use Cases:

Web app and mobile app development
Real-time analytics
Microservices and API integration

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Software as a Service (SaaS)

SaaS delivers software applications over the Internet, which are managed by the service provider.

- What You Get:
    - Fully managed applications
    - Accessible through web browsers or APIs
    - Automatic updates and maintenance
- User Responsibility:
    - Use of the application only
- Advantages:
    - No installation or infrastructure required
    - Accessible from any location and device
    - Subscription-based pricing

Software as a Service (SaaS) (Cont’d)

- **Examples:**
  - **Gmail**: Web-based email without any client installation.
  - **Dropbox**: Cloud file storage and sharing.
  - **Salesforce**: Customer Relationship Management (CRM) platform.
  - **Microsoft 365**: Cloud-based office productivity suite.
- **Use Cases:**
  - Email and communication tools
  - Customer management and support
  - File storage and collaboration

Comparison

Feature	IaaS	PaaS	SaaS
User Manages	OS, Middleware, Apps	Apps only	Nothing
Flexibility	High	Medium	Low
Developer-Friendly	Yes	Very High	No
Best For	System Admins, IT teams	Developers	End-users
Cost Model	Pay-per-use	Usage/Subscription	Subscription
Example	Amazon EC2, Azure VMs	Heroku, App Engine	Gmail, Dropbox